

Modal Assessment of Risk in Wireless Networks

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Abstract—A formalized treatment of modal distance in which metric structures are captured in a Kripke frame provides for a wide ranging semantics with which our intuitions concerning risk can be accurately modeled. In the model proposed here, we preserve the standard axes of *risk level* and *impact level* – where modal distance is equated with the former – and introduce a security strength function to capture defensive investments, as well as a perspicaciousness function that orients modal distance with our expectations concerning risk (i.e., a low score implies low risk).

Keywords—*risk, cybersecurity, modal logic*

I. INTRODUCTION

The fields of Safety Studies and Cyber Security overlap methodologically in their respective affordances for processes related to risk management, and therefore share a fundamental concern with uncertainty. There is, however, a discernible gap in how the fields treat risk: whereas an empirically motivated rate of failure predominates in Safety Studies, a threat's immanence – as logically derived from the kill chain – is the primary driver in Cyber Security. For example, Odd Olsen observes in his introduction to a series of multidisciplinary essays on standardization and risk, contributors either hold to a definition of risk that expresses “the relations between probabilities and expected outcomes, including the uncertainty always embedded in risks,” or else combine “threats, values, and the vulnerability of the object...at stake, where probabilities are avoided.”¹ It would appear, then, that the concept of risk is under some tension between qualitative and quantitative framings of its semantic content. Sven Hansson recognizes five prevailing uses, the first two of which are qualitative:

1. An *unwanted event* that may or may not occur.
2. The *cause* of an unwanted event that may or may not occur.
3. The *probability* of an unwanted event that may or may not occur.
4. The *statistical expectation value* of an unwanted event that may or may not occur.
5. The fact that a decision is made under conditions of *known probabilities* (i.e., “decision under risk,” not “decision under uncertainty”).²

The third usage, as exemplified by a statement like “The risk that an aggressive driver is involved in a severe accident is about 60%,” can be traced to computational solutions to the expected value $E(X)$ of a random variable X that were first worked out in a series of letters between Blaise Pascal and Pierre de Fermat.³ The fourth usage, however, entered into risk analysis in the wake of an influential 1975 report

from the U.S. Nuclear Regulatory Commission on the risk of accidents at U.S. nuclear power plants, and is typically associated with the formula $risk = hazard\ likelihood \times severity$.⁴ The fifth describes the conditions under which a safety engineering task should be undertaken.

II. MODAL RISK ASSESSMENT

Related work initiated by Duncan Pritchard (2005; 2007; 2008; 2015; 2022) advances the argument that the formal, more technical definitions of risk fail to capture key aspects of our intuitions about risk. Rather, a modal account that proceeds from evaluation of the same double (*level of risk, significance of the undesirable event*) avoids the unsatisfactory conclusions concerning rationality in those circumstances in which: (i) a high level of risk coincides with a low probability, but (ii) one's judgement about these matters runs counter to what the probability would recommend. The lottery is frequently given as an example in this literature, viz., whereas the probability of one's ticket being the winner is unequivocally low, the impact of a win is high. Pritchard argues that one's judgment departs from probability in this case because of the “easy possibility” that one's ticket wins, which is to say that all that is necessary to win is the correct configuration of different colored balls. In more formal terms, the circumstances under which one wins the lottery are *modally close* since, all things held equal, very little differs between the circumstances under which one loses (at least one ball does not match) and those under which one wins (all the balls match). Thus, Pritchard claims that while, in general, high probability events are modally close and low probability events are modally distant, where modal distance and probability depart from each other “one's judgement will tend to concern the modal closeness of the risk event in question rather than its probability.”⁵

For this reason, a modal account of risk can also explain why one would be concerned with checks and safeguards when serious risk is immanent. Consider the implementation of a safety system for a critical infrastructure: the probability of a failure of the safety system is low, but a disaster could be high risk if no safeguard were in place – indeed, such an event would be modally close since a failure of a single component would suffice to cause a failure. Hence, the addition of checks and safeguards to the safety system has the effect of making failure both more unlikely and modally more distant, thereby lowering risk. An equivalent analogy could be drawn for the cyber security domain: consider the hardware implementation of a key storage module. Whereas the creation of a trusted zone imposes a form of access control, the integrity of the keys remains at risk via attacks on the address and data buses that may connect the trusted zone to other nodes on the network. Thus, encryption and

¹ O. Olsen (2020). “The standardization of risk governance.” In O. Olsen, K. Juhl, P. Lindoe, & O. Engen (Eds.), *Standardization and risk governance. A multi-disciplinary approach*. (Routledge): 2000. 1997-2008

² S. O. Hansson (2022). “Risk.” plato.stanford.edu/entries/risk/.

³ See A. Hald. (1990). *History of probability and statistics and their applications before 1750*.

⁴ N. Rasmussen, et al. (1975). “Reactory Safety Study,” *WASH-1400, US NRC*. Also see R. P. Rechard. (1999), “Historical relationship between performance assessment for radioactive waste disposal and other types of risk assessment.” *Risk analysis*, 19(5), pp. 763-807.

⁵ D. Pritchard. (2022). “In defence of the modal account of legal risk.” *Synthese*, Vol. 200.

decryption routines are often included in this architecture, along with tamper resistance mechanisms to decrease the likelihood of an attack and mitigate an attempt such that its success is modally more distant.

III. RISKINESS UNDER A MODAL DISTANCE METRIC

The argument for a modal approach, then, proceeds as follows: The long standing stochastic view of risk, as exemplified in the latter-day *expected value* definition favored by Safety Studies and related engineering fields, is incomplete. The modal account, which draws from model-theoretic developments in the field of modal logic, more persuasively describes our intuitions about possible events and therefore constitutes the more general domain. It also provides a compelling explanation for the decisions that follow from our considerations of possible events. This paper adopts these premises in order to derive a model of a cyber security system that captures the riskiness of its components. Such a model will depend on a formalization of the concept of modal distance that emerges in the work of Johan van Benthem (1989; 2010), and which later receives key completeness and decidability results (see, e.g., F. Wolter and M. Zakharyashev, 2003; D. Gabelaia, et al., 2007).⁶ Following this body of work, I will interpret the Kripke accessibility relation R as an edge set such that a Kripke frame can be treated as a directed graph. Then, by assigning weights to these edges in order to represent a degree of accessibility between components, I will construct a riskiness graph whose connections give rise to both qualitative expressions (possible attack paths) and quantitative expressions (degree of difficulty in moving between vertices). The advantages of this approach are twofold: a varied set of parameters and expressions with which intuitions about the system can be matched, and a comprehensive means of representing the system that allows for evaluation at different scales or from different starting points.

A. Expressivity of the Spatialized Modal Distance Metric

In order to capture risk in terms of modal distance, an alethic relational regime is needed to constrain access (possibility; possible relations) with a set of logical conditions (propositions), and a metric regime is needed to ascribe weight (distance) and capability to each relation. These needs are served by representing the latter within a Kripke frame by means of the quintuple $\langle W, R, C, V, E \rangle$ consisting of

- a nonempty set W of *worlds* $\{w_0, w_1, w_2, \dots\}$ that identify possible states (*access*), where $w_0 \in W$ represents the current state,
- a function $R: \text{Ag} \rightarrow \mathcal{P}(W \times W)$ that assigns to each entity $a \in \text{Ag}$ a *possibility relation* (or *access channel*) $R_a \subseteq W \times W$, by virtue of which it is

- possible to denote with $wR_a v$ that entity a can, in its current state w , induce state v ,
- a function $C: \text{Ag} \rightarrow \mathbb{R}$ that assigns to every entity a a degree of knowledge about the state of its network,
- a function $V: W \rightarrow \mathcal{P}(\text{Prop})$ that assigns to every state a set of propositional atoms that are true in that state, where Prop is a countable set of propositional atoms, and
- a function $E: \bigcup_a R_a \subseteq W \times W \rightarrow \mathbb{R}^+$ that assigns to every pair of states a positive real number, and so (as a subset of the network $E \subseteq W \times W$) expresses the degree of distance between states.

Given such a structure it is possible to construct a single-layer distance graph, a directed weighted network $G = (W, E)$ that is composed of states whose connections to other states are given by E , a triple $(u, v, \text{Ag}(u, v))$ which carries access channel information encoded in R_a as an attribute. That is,

$$E = \{(u, v, \text{Ag} = \{a | (u, v) \in R_a\})\}, \quad (1)$$

where $(u, v) \in E$ is true only if at least one entity can induce v from u . Weights r are derived from C and computed as a function of pairs (u, v) :

$$r(u, v) = f(\{C(a) | (u, v) \in R_a\}). \quad (2)$$

We therefore select an appropriate function $g: \text{range}(C) \rightarrow \mathbb{R}$ that maps knowledge onto positive costs, and finally induce our desired metric $d: W \times W \rightarrow \mathbb{R}_{\geq 0}$. Then, the distance between two states $d(u, v)$ can be defined as the infimum from the range of possible sums – or paths – between states u and v , or

$$d(u, v) = \inf\{\sum_{i=0}^{n-1} r(x_i, x_{i+1}) \mid (x_i, x_{i+1}) \in E\}, \quad (3)$$

for $0 \leq i \leq n$. As a metric, we must also count on our distance to exhibit the following properties:

- $d(u, u) = 0$,
- $d(t, u) \leq d(t, u) + d(u, v)$ (triangle inequality), and
- $d(u, v) = d(v, u)$ (symmetry).

⁶ A comprehensive literature review is out of the scope of this paper, but the formal properties of modal distance that will be referenced here rests on work that draws from structural similarities between Kripke structures – relational models of possibility and necessity – on the one hand, and weighted metric or distance graphs – mappings of vertex pairs to real numbers (zero or positive), that quantify the distance between those pairs – on the other hand. These two fields emerge independently at roughly the same time in the 1950s, and gradually converge through: work by

Montague (1970) that generalizes Kripke frames to neighborhood semantics, and the development of fuzzy logic and graded modal logics by L. Zadeh (1965), P. Hajek (1998), and R. Fagin and J. Halpern (1994). For details, see M. Aiello, I. Pratt-Hartmann, J. van Benthem. (2007). *Handbook of Spatial Logics*, especially “What is Spatial Logic,” pp. 1-11 and “Modal Logics of Space,” pp. 217-98.

B. Risk as Modal Distance

Given our distance function between states, we may then express the riskiness of the process that is captured by that states pair in terms of this metric. As discussed in Section 2, risk occupies two dimensions – level of risk and impact level – and, accordingly, we would expect that our model captures both of these independent factors. So far, the distance expression, which we would like to equate with level of risk, is an adequate representation of an attack surface. If, for instance, “ $d(u, v)$ is small” is interpreted to signal high risk, by analogy this would imply that access between the states pair (u, v) is relatively unimpeded. More abstractly, we could say that the perspicaciousness of the process $uR_a v$, or the degree of information made available by the process – which is given by $E(u, v)$, is high and therefore presents a relatively large attack surface.

For a concrete example, let us examine the case of a simple 802.11 frame exchange, from probe request and response through authentication. Let us first assume an open system in which a wireless module STA moves from an idle state w_0 to a passive probing mode and sends a probe request (w_1). An access point AP sends a probe response (w_2), upon which STA initiates authentication with an authentication request, subsequently receives an authentication response, and finally associates with AP (w_3). Given such a scenario, we can populate the model as follows:

- $W = \{w_0, w_1, w_2, w_3\}$, where
 - w_0 : Idle state (“actual world”)
 - w_1 : STA sends probe request
 - w_2 : AP sends probe response
 - w_3 : STA associates (data link established)
- $Ag = \{a_{STA}, a_{AP}\}$
- For relation functions $R_{aSTA}, R_{aAP} \in W \times W$, we have edges
 - $(w_0 \rightarrow w_1) \in R_{aSTA}$: Probe sent
 - $(w_1 \rightarrow w_2) \in R_{aAP}$: Probe response
 - $(w_2 \rightarrow w_3) \in R_{aSTA}$: Association

This can be represented in a Kripke model, wherein states are represented by circles, and edges correspond to arrows that are labeled with the appropriate agent. Note that the actual world appears as a double circle (see Figure 1).

As discussed, in order to assess distance we need to assign weights by means of a function $g(C(a))$, where $C(a)$ is a scalar that represents an agent’s knowledge about the system in which it operates. The function g then represents an appropriate transformation on this value to reflect the incursion of some cost on the pathway between states. Let us assume, in this case, that the AP possesses the lion’s share of the knowledge about the network, and that the STA knows only its own identifying information, so that $C(AP) = 0.9$ and $C(STA) = 0.1$. Then, let $g(c) = r(u, v) = 1 - c$ to reflect the cost of knowledge; so, for

we have weights $r(w_0, w_1) = 1 - 0.1 = 0.9$,
 $r(w_1, w_2) = 1 - 0.9 = 0.1$, and $r(w_2, w_3) = 0.9$.

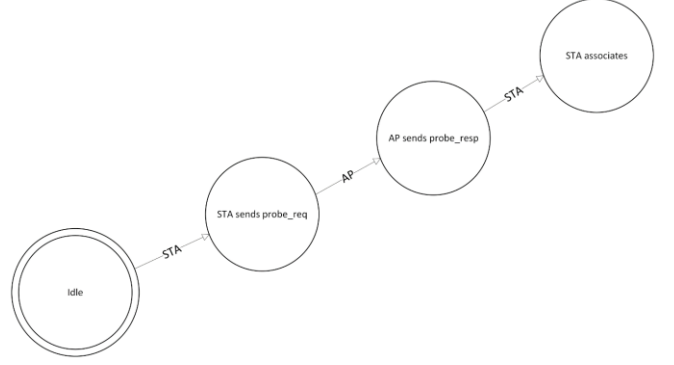


Figure 1: Kripke model showing worlds and agentic associations.

Since we have only one possible path,

$$d(w_0, w_3) = 0.9 + 0.1 + 0.9 = 1.9.$$

The significance of this value is that of a baseline, against which we can test whether the model can accommodate the addition of security measures such that distance is increased where those measures are present. To do so, let us model the Extensible Authentication Protocol (EAP) frame exchange and compare the results.

C. Security Measures as a Distance Enhancement

Since the EAP frame exchange includes the open system states, we will begin with these same states and the same assumptions about the STA and the AP. Once the STA is associated, however, the AP routes EAP frames encapsulated in RADIUS packets to a RADIUS server to conduct an authentication procedure that will involve the generation of a data encryption key – in this case, a pairwise master key (PMK) – followed by the four-way handshake. The full exchange is given in Figure 2. The STA initiates the exchange with the EAPOL start message (w_4), which begins a series of EAP request/response transmissions to establish the identities of the STA, the AP, and the RADIUS server (AS). The AP first requests the identity of the STA (w_5), to which the STA responds (w_6) and the AP subsequently turns to the AS to forward the response to it with an EAP access request appended (w_7). The AS then sends materials for authentication and en/decryption to the AP (w_8), which is to be forwarded to the STA (w_9). The STA responds in kind (w_{10}), and has its response forwarded to the AS (w_{11}). After validation of this message by the AS, the AS will send the result (Access or Reject) to the AP, resulting in a branching. If rejected (subset $w_{11.0.x}$), the transaction ends. If access is granted ($w_{11.1.0}$), the AP will

$$E = \{(w_0, w_1, R_{aSTA}), (w_1, w_2, R_{aAP}), (w_2, w_3, R_{aSTA})\}, \quad (4)$$

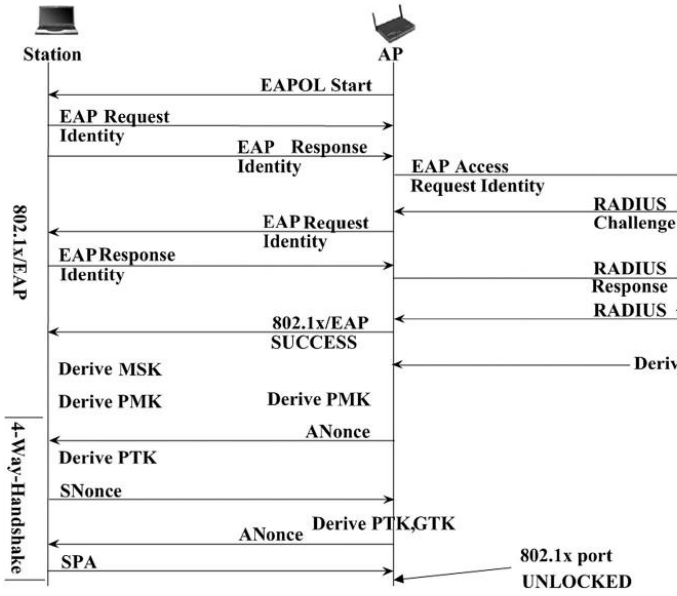


Figure 2: The EAP frame exchange following initial request-response.

then send the appropriate EAP (Success or Failure) message to the STA ($w_{11.1.1}, (w_{11.1.2})$).

Now, the PMK is established among the three parties, and the STA and the AP enter the four-way handshake. The AP generates the anonce and sends this to the STA ($w_{11.1.3}$), to which the STA responds with the snonce, the pairwise transmission key (PTK), and a message integrity code (MIC) to validate their mutual PMK ($w_{11.1.4}$). The AP will then load the PTK into its memory and request that the STA do the same ($w_{11.1.5}$); the STA ends the handshake when it completes this task and sends an ACK to the AP ($w_{11.1.6}$). This suggests

- $W = \{w_0, \dots, w_{11.1.4}, w_{11.1.5}, w_{11.1.6}\}$, where w_0 through w_3 are as before and there is a branching at w_{11} due to the possibility for a determination by the AS to reject (subset $w_{11.0.x}$) or permit (subset $w_{11.1.x}$) STA access.⁷
- $\text{Ag} = \{a_{STA}, a_{AP}, a_{AS}\}$
- For relation functions $R_{aSTA}, R_{aAP}, R_{aAS} \in W \times W$,

$$R_{aSTA} = \{(w_0 \rightarrow w_1), (w_2 \rightarrow w_3), (w_3 \rightarrow w_4), (w_5 \rightarrow w_6), (w_9 \rightarrow w_{10}), (w_{11.1.1} \rightarrow w_{11.1.2}), (w_{11.1.3} \rightarrow w_{11.1.4}), (w_{11.1.5} \rightarrow w_{11.1.6}), \}$$

$$R_{aAP} = \{(w_1 \rightarrow w_2), (w_4 \rightarrow w_5), (w_6 \rightarrow w_7), (w_{10} \rightarrow w_{11}), (w_{11.0.0} \rightarrow w_{11.0.1}), (w_{11.1.0} \rightarrow w_{11.1.1}), (w_{11.1.2} \rightarrow w_{11.1.3}), (w_{11.1.4} \rightarrow w_{11.1.5})\}$$

$$R_{aAS} = \{(w_7 \rightarrow w_8), (w_{11} \rightarrow w_{11.0.0}), (w_{11} \rightarrow w_{11.1.0})\}$$

Figure A in the Appendix provides a Kripke model of this extended scenario. To compute distance, we use our previous capability values $C(AP) = 0.9$ and $C(STA) = 0.1$, and let $C(AS) = 0.5$ under the presumption that the AS is an intermediate asset. We may also leave our function g unchanged. Then, as we are confronted with two distinct paths, it perhaps suffices to check the shorter of the two, in which the AS rejects the STA:

$$d(w_0, w_{11.0.2}) = 1.9 + \dots + 0.5 + 0.1 + 0.9 = 7.0.$$

It is quite clear that the mere addition of states was all that was needed to increase the distance. This matches our intuitions about the basic process of implementing security measures, and yet it also fails to discern whether a state is malevolent since the addition of an attacker's routine would similarly increase distance. Conceivably, this could be handled with an epistemic relation $K_a \subseteq W \times W$, which allows for a formulation of the form $wK_a v$ to express an agent a 's ability to distinguish worlds w and v . Specifically, $wK_a v$ holds when agent a 's degree of uncertainty is such that it cannot distinguish world w from world v , which is to say that a possesses insufficient knowledge to "know" w from v and so the two appear identical to a . If we introduce an attacker $a_{att} \in \text{Ag}$ to our scenario – which we extend, say, with a series of data transmissions ($w_{11.1.7.x} \rightarrow w_{11.1.7.y}$) following STA's ACK frame – $w_{11.1.7.i}K_a w_{11.1.7.j}$ implies that, given some ciphertext block m_i due to the PTK, which we denote by k , a_{att} is unable to distinguish $\{m\}_k$ in $w_{11.1.7.i}$ from $\{m\}_k$ in $w_{11.1.7.j}$. Conversely, if a_{att} possesses k , the content of m is no longer opaque and $w_{11.1.7.i}K_a w_{11.1.7.j}$ no longer holds since a_{att} now has grounds for differentiating $w_{11.1.7.i}$ from $w_{11.1.7.j}$.

It is important to observe that K_a was the primary extension of these spatialized modal models.⁸ In particular, it preserves the intuition behind modal distance, according to which worlds are enumerated according to their essential (modal) properties. In the context of our scenario, for example, the two worlds at issue differ only in the encrypted blocks due to k , as desired. However, risk, as we have conceived it, requires an assessment along both the axes of distance and impact. The model, therefore, is incomplete.

⁷ A number of other branchings due to error or failure are conceivable, but have been left unexplored for simplicity's sake.

⁸ See O. Kutz, "Logics of distance," Ph. D dissertation, Dept. of Comp. Sci., Univ. of Liverpool, Liverpool. 2004. [Online]. Available: <https://www.inf.unibz.it/~okutz/resources/EconnsandDistances.pdf>

In the next section, I will introduce two functions: a security strength function $S(u, v)$, which assesses a defensive score for measures like encryption or segmentation at either the edge or process level, and an impact level function $I(u)$, which uses a range of values, e.g., $[1, 10]$, to capture the severity of a CIA breach at a particular state or within a specific asset. In this way, the system can be approached as something more than a simple concatenation of state transitions. By means of S we will be able to affect some linear transformation on the paths through the system to reflect their level of security, and, by means of I , introduce an amplitude to the result that properly reflects the stakes or perceived value of the information that is transmitted. Finally, in order to orient the distance value with our sense of risk (so that a low result can be equated with low risk), we will use a mapping function $L(v)$ that yields a score L that *decreases* with distance $d(u, v)$. This gives us a risk expression that can be evaluated at some terminal or target state v (and which is still assessed from w_0).

D. Risk as the Product of Impact and Modal Distance

Let S be some weight function applied across some state transition $E(u, v, \text{Ag}(u, v))$ or at some state $w \in W$ to give a positive scalar indicating *security strength*, or:

$$S_{\text{process level}}: E \rightarrow \mathbb{R}_{\geq 0} \quad (5)$$

and

$$S_{\text{state level}}: W \rightarrow \mathbb{R}_{\geq 0}. \quad (6)$$

Then, choose some function h that represents the *security effect* of S at (u, v) , written $h(S(u, v))$, so that the total weight from u to v becomes

$$r(u, v) = (g(C(a)) | a \in \text{Ag}) \cdot h(S(u, v)) \quad (7)$$

Similarly, let I indicate the *impact* at some target state v , such that we can assess the intrinsic effect of a breach at v , independently of S :

$$I_{\text{eff}}(v) = I(v). \quad (8)$$

Next, let L be some mapping of the function $d(u, v)$ that yields a *perspicaciousness score*, given by

$$L(v) = (f(d(u, v), \alpha) | \alpha \in \mathbb{R}^+) \quad (9)$$

where α is some appropriate scaling factor on d . Risk at v is therefore a function $\text{Risk}(I, L)$ of the weighted and scaled path $u \rightarrow v$, $u, v \in W$, in system $W \times W$, which will be defined as:

$$\text{Risk}(v) = I(v) \cdot L(v). \quad (10)$$

IV. APPLICATION OF THE MODEL: SECURE REMOTE VEHICLE DIAGNOSTICS

For a full demonstration of the model, let us consider a mock-up of a secure wireless module that transmits key vehicle diagnostics from a simulated OBD-II interface to a remote server. On one end, we have a vehicle unit, represented by a Raspberry Pi B+ with an ARM 1176JZF-S processor that is coupled with a PiCAN FD board. On this module, a collection of Python libraries should be found that implement:

- basic inter-process communication (IPC) and data exchange (socket, json)
- a lightweight TLS handshake for wireless communication (mTLS, ssl)
- CAN frame exchange (can, can_obd)
- certificate authentication, key generation, and encryption (Fernet)
- low energy Bluetooth (BLE) activation (bluetooth)

This module would be responsible for:

- a connection to the OBD-II interface
- periodic data reads
- payload formatting
- frame encryption and transmission

On another end, we have the remote client represented by a second Raspberry Pi B+ - PiCAN FD board coupling with an identical set of Python libraries. This module would be responsible for:

- payload decryption and validation
- secure formatting of results for logging or display

A diagram of this set-up is provided below in Figure 3.

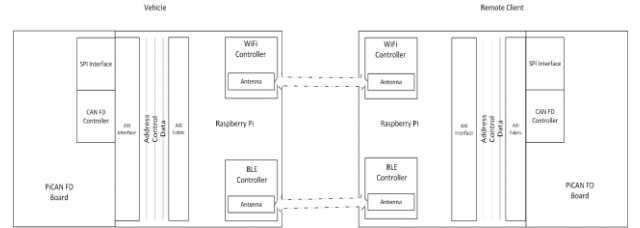


Figure 3: Remote vehicle diagnostic communication

Let us begin the enumeration of states with the observation that this two-module system should operate in a real-time environment. So, in choosing an actual state, let us presume that the vehicle side is expected to update the remote server on a periodic basis, beginning with a read command *get_obd_all()* that is read at the CAN FD Controller. There, we will introduce a branching to allow for a CAN frame check; if an error arises, let us assume that the path ends. Otherwise, let us assume that the Raspberry Pi induces a passing of the frame along the AXI bus to an L1 data cache, where it will wait to be routed either to the WiFi controller or the BLE controller. Since WiFi is the farther ranging transmission method, we will allow this controller to act as the default – in which case, the transmission proceeds as outlined in the previous section and terminates at state w_2 .

If a Bluetooth pairing has been signaled, i.e., a link layer connection has been established, we assume an elliptical curve Diffie-Hellman (ECDH) key exchange followed by an encrypted data transmission that similarly terminates at w_2 . A Kripke model of this scenario can be found in Figure B in the Appendix. The model elements and valuation choices are given below:

- $W =$
 $\{w_0, w_1, \{w_{1.0.0}\}, \{w_{1.1.0}\}, \{w_{1.1.1.0}\}, \{w_{1.1.1.1}\}, w_2\}$
with subsets
 - $\{w_{1.0.0}\}$: frame verification error branch,
 - $\{w_{1.1.1.0}\}$: WiFi transmission branch, and
 - $\{w_{1.1.1.1}\}$: BLE transmission branch.
- $Ag = \{\{\text{Module}_1\}, \{\text{Module}_2\}\}$, each with members
 - a_{CANFD_i}
 - a_{PiMem_i}
 - a_{PiWiFi_i}
 - a_{PiBLE_i}
- For $S_{process\ level}: E \rightarrow \mathbb{R}_{\geq 0}$, let $h(S(u, v)) = 1 + S(u, v)$, and
 - $S(w_0, w_1) = 0$ (no assumed OBD-II tamper protection or frame encryption)
 - $S(w_{1.1.0}, w_{1.1.1}) = 0$ (no assumed TrustZone access control)
 - $S(\{w_{1.1.1.0}\}, w_2) = 2$ (standard 802.11 protections)
 - $S(\{w_{1.1.1.1}\}, w_2) = 2$ (standard BLE protections)
- Since no protections have been implemented on the port or the memory components, we will assign a flat knowledge cost across all members of a given module and recognize a distinction only between the modules. So, for $g(C(a))$, where $g(C) = 1 - C$, let
 - $C(\{\text{Module}_1\})$: 0.5 (the vehicle contains sensitive CAN data)
 - $C(\{\text{Module}_2\})$: 0.2 (the client knows what it receives from the vehicle)
- Let $I = \{1, 10\}$, assigned over the following path segments of interest:
 - $\{w_0, w_1\}, \{w_{1.0.0}, w_{1.1.1}\}, \{w_{1.1.1.0}\}, \{w_{1.1.1.1}\}, w_2$: 10
 - $\{w_{1.0.0}\}$: 1
- Finally, let us define $L(v) = e^{-d(w_0, v)}$

V. CONCLUSION

Some questions arise, here. First, in assessing distance, we can observe the behaviors of our parameters on the two paths ($w_{1.1.1.0} \rightarrow w_2$) and ($w_{1.1.1.1} \rightarrow w_2$) leading to the data transmission state w_2 . In this case, there is tension between fidelity to the protocols we are evaluating and the intuitions we wish to test. On the one hand, both channels have standard protections implemented, with little to regard in the way of comparative risk. On the other hand, since

they constitute alternative channels of transmission, our model puts their relative number of states under consideration such that the leaner of the two is granted an advantage in being selected as the representative path. Therefore, if this were undesirable, a choice must be made to either populate the model with worlds in such a way that the two channels are of equal size, or else, favoring realism, allow the security strength valuation to account for this discrepancy with a hyperparameter β , which could be introduced in our function h , e.g.,

$$1 + \beta S(u, v). \quad (11)$$

Second, in evaluating impact, we may want to provide for situations in which some security measure mitigates the cost of an attack that was successful by some other means. For example, a tamper resistant hard drive may not prevent theft, but, in case it were stolen, it would confound efforts to gain access to its contents and therefore lessen the cost. In considering such a scenario, we may wish to employ an expression similar to the one below:

$$I_{\text{eff}}(v) = I(v) \cdot (1 - m(S_{\text{state level}}(v))) \quad (12)$$

where $m(S) = \gamma \frac{S}{1+S}$. Finally, automation of the world enumeration and valuation selection tasks is computationally feasible. Implementation of this encoding of the model in a language such as Python would facilitate its coupling with deep learning networks like a multi-layer perceptron, which would in turn allow for fine tuning of the hyperparameters in the functions for security strength, impact, and distance.

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APPENDIX

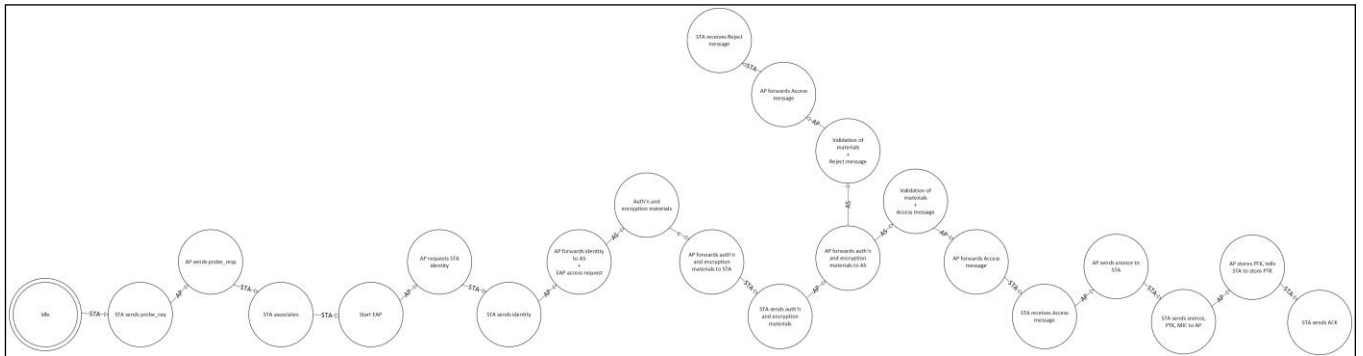


Figure A: Extended Kripke model of EAP-TLS frame exchange

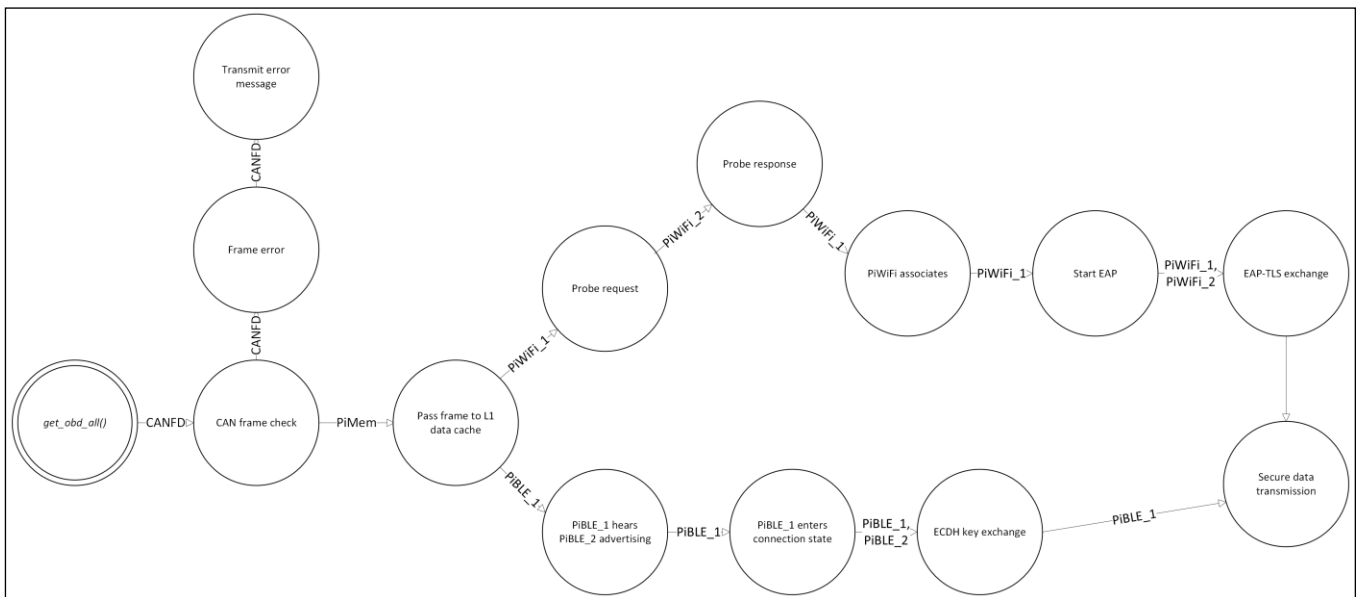


Figure B: Kripke model of secure vehicle diagnostics